

Daily Macro Brief Agent - 1-Page Memo

Design Tradeoffs

The prototype is built as a local-first agent that can be run manually, by GitHub workflow dispatch, or by a scheduler on an always-on machine. I chose Gmail SMTP rather than Telegram because email fits the intended daily workflow and requires no new user habit. Gemini 2.5 Flash-Lite is the default LLM because the model's job is narrow: synthesize structured inputs into concise macro prose. Code, not the LLM, owns market data, tables, charting, source links, validation, logging, and delivery.

The core design choice is to separate facts from writing. Market numbers come from public APIs or scraping: Yahoo Finance chart data, Japan MOF JGB data, Frankfurter FX reference rates, and CoinGecko BTC data. If live market data is unavailable and no cached real row exists, the value cells are left blank rather than filled with generated numbers. Calendar rows come from the Fair Economy / Forex Factory weekly feed; RSS sources feed Theme Radar. The LLM receives only structured facts and is validated for JSON shape, word limits, market-number consistency, source reuse, and portfolio logic before output is accepted.

The brief is tailored to a macro PM rather than a news reader. The dashboard is a table with “Reading” rather than generic commentary. The 3 Things section is exactly three items with portfolio implications. The chart is USD/JPY because the assumed book is long USD/JPY; the current chart uses about three months of history and highlights the latest five observations. Theme Radar uses curated RSS sources with source-depth labels, so readers can see when the agent is using feed-level text rather than full article text. Scheduling was tested honestly: GitHub manual send worked, GitHub scheduled triggers were unreliable in short-window tests, and a MacBook `launchd` scheduled send succeeded end to end.

Position And Theme Assumptions

Because the PDF does not give the firm's real book, the prototype uses explicit assumptions stored in `inputs/portfolio/positions`. The assumed book is long USD/JPY, overweight gold, and exposed to an EM debt basket. The assumed house themes are higher-for-longer rates, fiscal pressure and term premium, China demand weakness, and policy divergence across major central banks. Position rows are effective-date updates, so the latest prior row carries forward until changed or closed.

Three V2 Features Not Completed

1. Add redundant or paid market/calendar providers so the agent is less dependent on free public feeds.
2. Expand Theme Radar from RSS-level text to permission-aware full-text reading for research notes, speeches, and selected newsletters.
3. Load feedback history from the questionnaire into ranking and prompt construction so the PM can teach the agent what is useful over time.

One-Month Full-Time Roadmap

With one month full-time, I would turn this into a production briefing system: add robust provider redundancy, source-health monitoring, schedule alerting, richer article ingestion, feedback-driven ranking, and a cleaner deployment target such as a VPS or cloud scheduler. I would also add a daily evaluation report showing source freshness, failed fetches, token use, runtime, and whether the delivered brief met all formatting and factual guardrails.

Actual Hours Spent

To be filled honestly by Tong Xu before submission: _____ hours.